

INVERGORDON HARBOUR, United Kingdom

WMO: 030580

Lat: 57.683N

Lon: 4.167W

Elev: 3

StdP: 101.29

Time Zone: 0.00 (EUW)

Period: 90-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-2.6	-1.3	-4.9	2.5	0.1	-3.9	2.7	1.0	14.9	7.2	13.4	7.9	1.6	280	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.6	21.0	16.4	19.5	15.6	18.3	14.9	17.2	19.9	16.2	18.7	15.4	17.8	4.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	11.3	18.5	14.9	10.6	17.7	14.1	10.0	16.9	48.1	20.2	45.3	18.8	43.0	17.9	20.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.3	11.5	10.0	DB	-4.7	23.8	2.7	2.2	-6.7	25.4	-8.2	26.6	-9.8	27.9	-11.8	29.5	(4)
(5)			WB	-5.0	18.6	2.6	0.9	-6.9	19.3	-8.4	19.9	-9.9	20.4	-11.8	21.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	4.3	4.9	6.1	7.4	10.2	12.4	14.5	14.8	12.5	9.4	6.4	4.2
	DBStd	4.52	2.49	2.90	2.49	2.43	2.58	2.09	1.91	2.09	2.15	2.48	2.91	3.03
	HDD10.0	896	177	146	122	84	28	4	0	0	4	41	110	182
	HDD18.3	3427	436	377	378	329	252	177	119	111	175	278	357	439
	CDD10.0	512	0	1	2	5	34	77	140	149	79	21	3	1
	CDD18.3	3	0	0	0	0	0	0	1	2	0	0	0	0
	CDH23.3	6	0	0	0	0	0	0	3	3	1	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.2	4.6	5.5	5.2	4.5	3.9	3.9	3.7	3.5	3.7	4.2	4.0	4.3
Precipitation	PrecAvg	885	89	81	75	62	58	67	51	59	78	91	87	81
	PrecMax	1157	175	165	164	108	120	133	93	122	246	152	124	146
	PrecMin	669	13	26	34	34	28	25	24	21	32	23	18	9
	PrecStd	124	47	42	39	21	26	33	20	27	49	34	28	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	12.6	12.8	14.8	19.5	20.9	23.4	22.8	21.2	17.4	14.4	12.2
		MCWB	9.3	9.8	9.5	10.7	14.1	14.7	17.6	17.2	16.6	13.7	11.8	10.3
	2%	DB	10.3	11.8	11.7	13.4	17.5	18.6	20.4	21.3	18.4	15.3	13.1	11.0
		MCWB	8.0	9.4	8.9	9.8	12.9	14.1	16.0	16.9	15.0	12.5	10.8	9.0
	5%	DB	9.3	10.9	10.9	12.3	15.8	17.3	19.0	19.7	17.0	14.1	11.7	10.2
		MCWB	7.2	8.7	8.4	8.9	11.9	13.1	15.6	16.1	13.9	11.5	9.6	8.2
	10%	DB	8.1	9.7	10.0	11.3	14.3	16.0	17.9	18.3	15.9	13.0	10.7	8.7
		MCWB	6.3	7.6	7.8	8.5	10.9	12.2	15.1	15.2	13.1	10.6	8.8	7.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	10.6	10.2	11.4	14.6	15.8	18.0	18.9	16.8	14.0	12.1	10.5
		MCDB	11.5	12.2	12.2	14.2	19.0	19.1	21.8	21.3	19.7	17.0	14.0	11.9
	2%	WB	8.3	9.7	9.3	10.2	13.4	14.7	16.9	17.6	15.4	12.9	11.0	9.1
		MCDB	10.0	11.5	11.4	12.7	17.0	18.0	19.4	20.0	17.8	15.1	13.0	10.6
	5%	WB	7.3	8.7	8.7	9.5	12.2	13.7	16.1	16.5	14.4	11.9	9.8	8.3
		MCDB	9.1	10.7	10.7	11.7	15.1	16.7	18.5	19.2	16.4	13.8	11.4	10.0
	10%	WB	6.4	7.5	7.8	8.8	11.2	12.8	15.3	15.5	13.6	11.0	9.0	7.1
		MCDB	8.0	9.7	9.8	10.8	13.8	15.3	17.5	18.0	15.3	12.7	10.6	8.7
Mean Daily Temperature Range	5% DB	MDBR	3.8	4.0	4.6	5.1	5.7	5.4	5.3	5.6	5.4	4.6	3.9	3.8
		MCDBR	5.0	4.2	5.4	6.6	8.1	7.5	7.0	7.5	7.1	6.0	4.9	5.3
		MCWBR	4.3	3.5	3.9	4.3	4.7	4.3	4.1	4.2	4.7	4.4	4.1	4.5
	5% WB	MCDBR	4.9	4.3	5.3	5.8	7.8	7.1	6.5	7.1	6.3	5.7	4.7	5.0
		MCWBR	4.3	3.8	4.2	4.2	4.8	4.4	4.2	4.5	4.6	4.6	4.1	4.4
Clear-Sky Solar Irradiance	taub	0.317	0.340	0.363	0.383	0.384	0.379	0.382	0.368	0.355	0.340	0.318	0.305	
	taud	2.202	2.283	2.288	2.277	2.325	2.372	2.377	2.422	2.456	2.404	2.257	2.144	
	Ebn at Noon	517	689	779	825	850	859	847	835	790	688	515	405	
	Edh at Noon	52	74	96	113	115	111	109	97	81	65	49	41	
All-Sky Solar Radiation	RadAvg	0.38	1.00	2.02	3.39	4.58	4.47	4.25	3.38	2.35	1.23	0.51	0.27	
	RadStd	0.05	0.12	0.18	0.27	0.32	0.41	0.39	0.21	0.19	0.12	0.05	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page